

EDUCATION

UC Berkeley

Ph.D. in Computer Science

Berkeley, USA

Aug 2025 – Present

McGill University

B.Sc. in Honours Computer Science

Montreal, Canada

Aug 2021 – Dec 2024

– Thesis: [Cell Morphology-Guided Small Molecule Generation with GFlowNets](#)

RESEARCH EXPERIENCE

UC Berkeley / Berkeley AI Research

Ph.D. Student – Advisor: Prof. Yun S. Song

Berkeley, USA

Aug 2025 – Present

- Leading various projects on deep probabilistic models of protein evolution and dynamics.
- Providing technical mentorship and overseeing projects for undergraduate students.

SickKids Research Institute

Research Technologist I – Advisor: Prof. Michał Koziarski

Toronto, Canada

Jan 2025 – Apr 2025

- Combined generative models with Bayesian optimization for batched design of synthesizable molecules.
- Developed SciGym, a benchmark for closed-loop LLM experiment design across simulated biological systems.

Mila - Quebec AI Institute

Student Researcher – Advisor: Prof. Jian Tang

Montreal, Canada

Aug 2024 – Jan 2025

- Contributed to EBA & ESMDiff, methods for physically calibrated and efficient protein conformation generation.

BigHat Biosciences

Machine Learning Intern – Supervisor: Dr. Hunter Elliott

San Mateo, USA

May 2024 – Aug 2024

- Developed fitness-conditional generative models for antibody variant design.

Mila - Quebec AI Institute

Student Researcher – Advisor: Dr. Michał Koziarski

Montreal, Canada

Jan 2024 – May 2024

- Developed generative flow networks (GFlowNets) for cell-morphology based small molecule discovery.
- Ran ablations of QGFN, which combines GFlowNets with Q-learning for high-reward, diverse sampling.

PREPRINTS AND PUBLICATIONS

- [1] **Lu, Stephen Zhewen***, A. Vermani*, K. Sanno, J. Lu, F. A. Matsen, M. Jagota, and Y. S. Song, “Conditionally site-independent neural evolution of antibody sequences,” in **ICML**, 2026.
- [2] H. Duan*, **Lu, Stephen Zhewen***, C. F. Harrigan, N. Desai, J. Lu, M. Koziarski, L. Cotta, and C. J. Maddison, “Measuring scientific capabilities of language models with a systems biology dry lab,” in **NeurIPS Datasets & Benchmarks Track**, 2025.
- [3] J. Lu, X. Chen, **Lu, Stephen Zhewen**, A. Lozano, V. Chenthamarakshan, P. Das, and J. Tang, “Aligning protein conformation ensemble generation with physical feedback,” in **ICML**, 2025.
- [4] J. Lu, X. Chen, **Lu, Stephen Zhewen**, C. Shi, H. Guo, Y. Bengio, and J. Tang, “Structure language models for protein conformation generation,” in **ICLR**, 2025.
- [5] **Lu, Stephen Zhewen**, J. Lu, H. Guo, and J. Tang, “Towards protein sequence & structure co-design with multi-modal language models,” in **ICLR LMRL Workshop**, 2025.
- [6] E. Lau, **Lu, Stephen Zhewen**, L. Pan, D. Precup, and E. Bengio, “Qgfn: Controllable greediness with action values,” in **NeurIPS**, 2024.
- [7] **Lu, Stephen Zhewen**, Z. Lu, E. Hajiramezanali, T. Biancalani, Y. Bengio, G. Scalia, and M. Koziarski, “Cell morphology-guided small molecule generation with gflownets,” in **ICML SPIGM Workshop**, 2024.

- [8] S. Nagaraj, V. Harish, L. G. McCoy, **Lu, Stephen Zhewen**, *et al.*, “From clinic to computer and back again: Practical considerations when designing and implementing machine learning solutions for pediatrics,” **Current Treatment Options in Pediatrics**, vol. 6, no. 4, pp. 336–349, 2020.

AWARDS

Y Combinator W26	2025
– Selected for the Winter 2026 batch; declined to pursue doctoral research.	
Laurie Hendren Memorial Scholarship	2024
– Awarded by the McGill Faculty of Science for academic excellence in Computer Science.	
8VC Fellowship	2024
– Sponsors students for internship at an 8VC portfolio company.	
Emily Ross Crawford Scholarship	2022
– Awarded to candidates of high academic merit in the McGill Faculty of Science.	
Schulich Leader Scholarship	2021 – 2024
– National STEM scholarship for top high school graduates demonstrating academic excellence and leadership.	
Canada Wide Science Fair Silver Prize	2019
– Awarded to 20 exceptional projects in each grade category among Canada’s brightest young minds.	
Hydro-Québec Super Science Fair Winner	2019
– Awarded to the most outstanding and innovative student scientific project at the provincial finals.	
Royal Conservatory of Music (RCM)	2017
– ARCT Performer’s Diploma in Piano.	

SERVICE

Delegate, Berkeley Graduate Assembly	2026
– Represent EECS concerns to the university-wide Graduate Assembly.	
Member, Berkeley EECS Social Committee	2026
– The social committee organizes events and holiday parties for Berkeley EECS PhD students.	

INVITED TALKS

BigHat Biosciences	August 2026
Denosing the Noise Journal Club	July 2026
Listgarten Group, UC Berkeley	July 2026
AstraZeneca AIXBio Frontiers Seminar Series	February 2026
McGill ML Reading Group	September 2025

REVIEWING

Transactions on Machine Learning Research (TMLR)	2026
Neural Information Processing Systems (NeurIPS)	2026
International Conference on Machine Learning (ICML)	2026
International Conference on Learning Representations (ICLR)	2026
ICLR Workshop on Learning Meaningful Representations of Life (LMRL)	2025
ICLR Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)	2025

MENTORSHIP

Yvette Chen – UC Berkeley	Undergrad
Kazuo Nakamura – UC Berkeley	Undergrad
Daniel Wu – SickKids Research Institute	Undergrad